Unit 4: Volatility Models (ARCH/GARCH)

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Applied Timeseries in R
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Outline

By the end of this unit you will be able to:

1. Describe the stylised facts of financial return volatility
2. Write down the GARCH(1,1) variance equation and interpret its parameters
3. Explain how GJR-GARCH and eGARCH capture the leverage effect
4. Fit and compare GARCH models in R using `rugarch`
5. Compute Value at Risk and Expected Shortfall from GARCH forecasts

Companion script
Open `Unit-4-Volatility-GARCH.R` — it simulates GARCH processes, fits multiple variants to real data and computes risk measures.

Stylised Facts of Financial Returns

Financial return series exhibit several **stylised facts**:

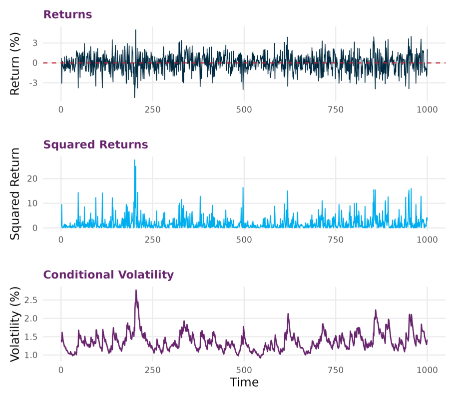
1. **Volatility clustering:** large changes followed by large changes
2. **Fat tails** (leptokurtosis): extreme returns more frequent than normal
3. **Leverage effect:** negative returns increase volatility more than positive ones
4. **Mean reversion:** high/low volatility is temporary

The gap ARIMA leaves

ARIMA captures the **conditional mean**.

GARCH captures the **conditional variance** — how the *spread* of returns evolves over time.

Volatility Clustering in Action



Top: returns with varying amplitude. Middle: squared returns reveal clusters. Bottom: the conditional volatility σ_t.

Detecting ARCH Effects

Visual evidence:

- Returns appear uncorrelated but **squared returns** show significant autocorrelation
- Excess kurtosis (kurtosis > 3)
- Jarque–Bera test rejects normality

Formal tests:

- Ljung–Box test on r_t^2: autocorrelation in squared returns
- Engle's ARCH-LM test: regress $\hat{\epsilon}_t^2$ on its lags

Key R functions

```
moments::kurtosis()
tseries::jarque.bera.test()
```

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The ARCH Model

ARCH(q) Model

Engle (1982)

The return process:

$$r_t = \mu_t + \epsilon_t, \qquad \epsilon_t = \sigma_t z_t, \qquad z_t \sim N(0,1)$$

The **ARCH(q)** conditional variance equation:

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \alpha_2 \epsilon_{t-2}^2 + \cdots + \alpha_q \epsilon_{t-q}^2$$

- Volatility depends on past squared shocks
- $\omega > 0$, $\alpha_i \geq 0$ ensure $\sigma_t^2 > 0$
- Large shocks $\Rightarrow$ high future volatility

Limitation

ARCH requires many lags to capture persistence. GARCH solves this with a single lag of σ_{t-1}^2.

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The GARCH Model

Standard GARCH(p, q)

Bollerslev (1986)

$$\sigma_t^2 = \omega + \sum_{j=1}^{q} \alpha_j \, \epsilon_{t-j}^2 + \sum_{i=1}^{p} \beta_i \, \sigma_{t-i}^2$$

GARCH(1,1) — the workhorse model:

$$\sigma_t^2 = \omega + \alpha_1 \, \epsilon_{t-1}^2 + \beta_1 \, \sigma_{t-1}^2$$

α_1	Reaction to new shocks (ARCH effect)
β_1	Persistence of past variance (GARCH effect)
$\alpha_1 + \beta_1$	Total persistence (close to 1 for equities)
Half-life	$-\ln 2 / \ln(\alpha_1 + \beta_1)$
Unconditional var.	$\bar{\sigma}^2 = \omega/(1 - \alpha_1 - \beta_1)$

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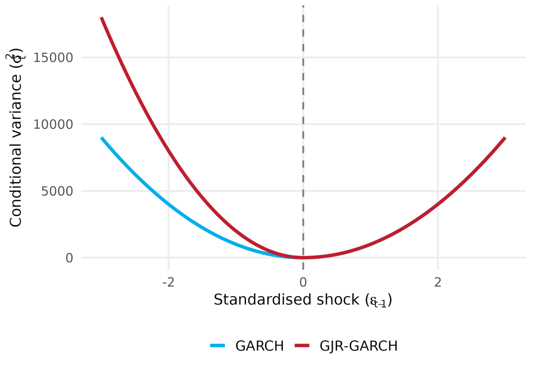
Asymmetric GARCH Models

The Leverage Effect



The **news impact curve** shows how a shock ϵ_{t-1} affects next-period variance. Standard GARCH is symmetric; GJR-GARCH adds extra impact for negative shocks (the “leverage effect” observed in equity markets).

GJR-GARCH (Threshold GARCH)

Glosten, Jagannathan & Runkle (1993)

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \gamma_1 \, I(\epsilon_{t-1} < 0) \, \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

- $I(\cdot)$: indicator function (1 if shock is negative)
- $\gamma_1 > 0 \Rightarrow$ negative shocks increase volatility more
- Total impact of negative shock: $\alpha_1 + \gamma_1$
- Total impact of positive shock: α_1

In `rugarch`

```
model = "gjrGARCH"
```

The **`gamma1`** parameter captures asymmetry. Test significance to determine if leverage is present.

Exponential GARCH (eGARCH)

Nelson (1991)

$$\ln(\sigma_t^2) = \omega + \alpha_1 z_{t-1} + \gamma_1(|z_{t-1}| - E|z_{t-1}|) + \beta_1 \ln(\sigma_{t-1}^2)$$

where $z_t = \epsilon_t/\sigma_t$ are standardised residuals.

Advantages:

- Models the **log** of variance $\Rightarrow$ always positive (no parameter restrictions)
- α_1: sign effect (asymmetry)
- γ_1: size effect

In `rugarch`

```
model = "eGARCH"
```

Often preferred in practice because it avoids non-negativity constraints.

Fitting GARCH Models in R

The `rugarch` Workflow

Five-step process:

1. Specify: `ugarchspec()`
2. Fit: `ugarchfit()`
3. Diagnose: `infocriteria()`, residuals
4. Forecast: `ugarchforecast()`
5. Simulate: `ugarchpath()`

Model string	Variant
`"sGARCH"`	Standard
`"gjrGARCH"`	GJR (threshold)
`"eGARCH"`	Exponential
`"apARCH"`	Asymmetric Power
`"iGARCH"`	Integrated

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Specifying and Fitting a Model

```
spec <- ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(1,1)),
  mean.model = list(armaOrder = c(1,0), include.mean = TRUE),
  distribution.model = "std")

fit <- ugarchfit(spec, data = returns)
```

Distribution choices:

- `"norm"` — Normal
- `"std"` — Student's *t* (fat tails)
- `"sstd"` — Skewed Student's *t*

Model comparison

`infocriteria(fit)[1:2]` — AIC, BIC

`persistence(fit)`

`-log(2)/log(persistence(fit))` — half-life

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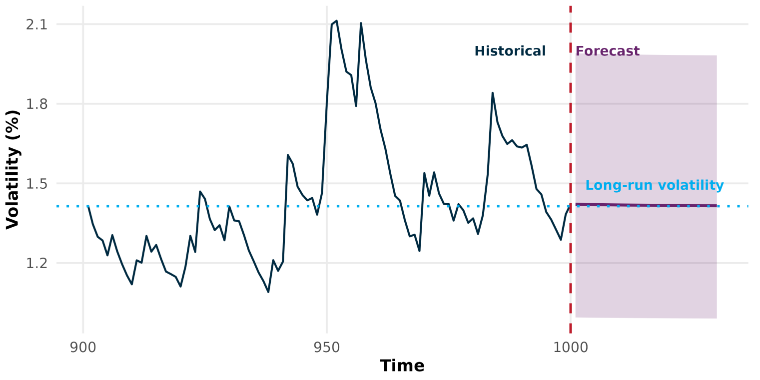
Forecasting Volatility & Risk

Volatility Forecasts



The forecast reverts toward the unconditional volatility $\bar{\sigma}$. The speed of reversion depends on persistence $(\alpha_1 + \beta_1)$: higher persistence $\Rightarrow$ slower reversion.

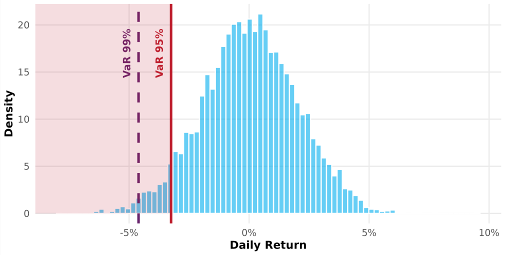
Value at Risk (VaR)

At confidence level $1 - \alpha$:

$$\text{VaR}_\alpha = \hat{\mu} + \hat{\sigma} \cdot q_\alpha(z)$$

where $q_\alpha(z)$ is the α-quantile of the standardised distribution.

- Answers: "What is the worst loss at the α probability level?"
- Typically $\alpha = 0.01$ or 0.05



Expected Shortfall (ES)

Expected Shortfall (CVaR): "Given that we are in the worst α% of cases, what is the expected loss?"

$$\text{ES}_\alpha = E[r_t \mid r_t < \text{VaR}_\alpha]$$

For Student's t with ν degrees of freedom:

$$\text{ES}_\alpha = \hat{\mu} - \hat{\sigma} \cdot \frac{f_\nu(q_\alpha)}{\alpha} \cdot \frac{\nu + q_\alpha^2}{\nu - 1}$$

Regulatory context

Basel III/IV requires banks to report both VaR and ES for market risk capital. ES is preferred because it is **coherent** (satisfies sub-additivity).

Exercises

Open `Unit-4-Volatility-GARCH.R` — the script provides simulation and fitting examples before each exercise.

1. **Simulate:** (a) Generate GARCH(1,1) paths with high persistence ($\alpha = 0.1, \beta = 0.88$) and low persistence ($\alpha = 0.3, \beta = 0.2$); (b) compare volatility plots
2. **Fit:** (a) Fit sGARCH(1,1) and GJR-GARCH(1,1) to S&P 500 returns; (b) is `gamma1` significant?
3. **Compare:** Use `infocriteria()` to rank the models by AIC/BIC. Which model wins?
4. **Risk:** (a) Forecast volatility 30 days ahead; (b) compute 1-day VaR and ES at 99% confidence

Hint: Use `ugarchspec()` with `distribution.model = "std"` for all models.

Summary

You can now:

- Identify volatility clustering, fat tails and leverage effects in return data
- Write down and interpret the GARCH(1,1) variance equation
- Explain how GJR-GARCH and eGARCH capture asymmetric responses
- Fit, compare and forecast GARCH models using `rugarch`
- Compute VaR and Expected Shortfall from GARCH volatility forecasts

Next: Unit 5 — Portfolio Optimisation & Risk Management